

Algebra 1 Unit 1 Assessment Guide

General Information	Unit 1 introduces students to the many functions they will be learning throughout the course. It is not expected that students have a thorough knowledge of these functions by the end of Unit 1. Finding values using function notation is an important skill for students to have gained from Unit 1. While this skill is introduced in this unit, it will be used throughout their studies in Algebra 1. Understanding the laws of exponents, finding values, and writing equivalent expressions using the laws of exponents is a culmination of students learning the laws of exponents throughout Grades 6-8. Applying the laws of exponents to write equivalent radical expressions is new to students in Algebra 1. If students are struggling with the basics of laws of exponents, consider using resources from Grade 7, Unit 8, Lessons 1-3 (Benchmark MA.7.NSO.1.1) and/or Grade 8, Unit 4, Lessons 1-6 (Benchmark MA.8.NSO.1.3) to provide remediation support for students. Operations with radicals is an application of students understanding the connection between the laws of exponents and radical expressions and is newly introduced in Algebra 1. This builds upon students' work with square roots and cube roots from Grade 8.
Calculator Information	The assessment is no calculator.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1a	Benchmark(s)	MA.912.F.1.1
Absolute Value Function		Recommended Scoring (4 Total Points)	Consider no partial credit.
Question	1b	Benchmark(s)	MA.912.F.1.1
Quadratic Function		Recommended Scoring (4 Total Points)	Consider no partial credit.
Question	2a	Benchmark(s)	MA.912.F.1.2
-11		Recommended Scoring (4 Total Points)	Consider no partial credit.
Question	2b	Benchmark(s)	MA.912.F.1.2
10		Recommended Scoring (4 Total Points)	Consider no partial credit.
Question	2c	Benchmark(s)	MA.912.F.1.2
-1		Recommended Scoring (4 Total Points)	Consider no partial credit.

Question	3a	Benchmark(s)	MA.912.F.1.2
1		Recommended Scoring (4 Total Points)	Consider no partial credit.
		Additional Notes	Monitor students whose answer is 4 as this is a common misconception.
Question	3b	Benchmark(s)	MA.912.F.1.2
-6		Recommended Scoring (4 Total Points)	Consider no partial credit.
		Additional Notes	Monitor students whose answer is -8 as this is a common misconception.
Question	3c	Benchmark(s)	MA.912.F.1.2
-5		Recommended Scoring (4 Total Points)	Consider no partial credit.
		Additional Notes	Monitor students whose answer is -4 as this is a common misconception.

Question	4a	Benchmark(s)	MA.912.F.1.2
$M(10) =$ 5079		Recommended Scoring (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	4b	Benchmark(s)	MA.912.F.1.2
Answers may vary. In 1980 U.S. passengers traveled 5079 millions of miles on a train.		Recommended Scoring (3 Total Points)	Consider giving students partial credit if they correctly interpret part of the answer. Students may rewrite as 5,079,000,000 miles.
		Additional Notes	Students will continue to learn about interpreting within a real-world context throughout the course.
Question	5	Benchmark(s)	MA.912.F.1.2
$P(2) =$ 9992		Recommended Scoring (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	6a	Benchmark(s)	MA.912.NSO.1.1
$x = 4$		Recommended Scoring (4 Total Points)	Please note that students could write 4 as $\frac{32}{8}$, $\frac{16}{4}$ or any other equivalent answer.
Question	6b	Benchmark(s)	MA.912.NSO.1.1
$y = 8$		Recommended Scoring (4 Total Points)	Consider no partial credit.
Question	6c	Benchmark(s)	MA.912.NSO.1.1
$k = 3$		Recommended Scoring (4 Total Points)	Consider no partial credit.
Question	6d	Benchmark(s)	MA.912.NSO.1.1
$n = -3$		Recommended Scoring (4 Total Points)	Please note that students could write -3 as $\frac{-15}{5}$ or any other equivalent answer.

Question	7	Benchmark(s)	MA.912.NSO.1.1, MA.912.NSO.1.4
	☐ Expression A ☐ Expression B ☐ Expression C ☐ Expression D ☐ Expression E ☐ Expression F	Recommended Scoring (12 Total Points)	Consider 3 points for each correct choice. If students choose an expression that is not equivalent, then consider deducting 3 points for each incorrect choice.
Question	8	Benchmark(s)	MA.912.NSO.1.1, MA.912.NSO.1.4
	☐ Expression A ☐ Expression B ☐ Expression C ☐ Expression D ☐ Expression E ☐ Expression F	Recommended Scoring (9 Total Points)	Consider 3 points for each correct choice. If students choose an expression that is not equivalent, then consider deducting 3 points for each incorrect choice.

Question		9a	Benchmark(s)	MA.912.NSO.1.1
Value 162			Recommended Scoring (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question		9b	Benchmark(s)	MA.912.NSO.1.1
Value $\frac{9}{2}$			Recommended Scoring (4 Total Points)	Students may provide any equivalent answer. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	10	Benchmark(s)	MA.912.NSO.1.4
$4\sqrt{3}$		Recommended Scoring (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	11a	Benchmark(s)	MA.912.NSO.1.4
Value	12	Recommended Scoring (4 Total Points)	Consider no partial credit.
Question	11b	Benchmark(s)	MA.912.NSO.1.4
Value	$32\sqrt{2}$	Recommended Scoring (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Please note that students could write $32\sqrt{2}$ as $16\sqrt{8}$.